First, $19^{659} \equiv 1^{659} \equiv 1 \pmod{2}$. Next, $19^{659} \equiv 1^{659} \equiv 1 \pmod{3}$. Finally, $19^{659} \equiv 4^{659} \equiv (-1)^{659} \equiv -1 \equiv 4 \pmod{5}$. To finish the problem off, we need to find a number that is 1 (mod 2), 1 (mod 3), and 4 (mod 5). We could do this systematically, but it's easier not to. By inspection, 4 is a number that is 1 (mod 3) and 4 (mod 5), and any other such number is congruent to 4 (mod 15). That leaves 4 and 19 as possibilities. Since only 19 is 1 (mod 2), that's the answer.

The point here is that we can solve this problem not by working modulo a large number directly, but by working with smaller numbers, where it's easier to see what's going on. In fact, we may always restrict to the case where congruences are modulo prime powers. For instance, when working modulo $120 = 2^3 \times 3 \times 5$, it suffices to work modulo 8, 3, and 5.

Here is another thing we can do with the Chinese Remainder Theorem:

Example. Find all integer solutions to $x^3 - x + 1 \equiv 0 \pmod{35}$. Let us write $f(x)$ for $x^3 - x + 1$. To solve this, we note that the statement $y \equiv 0 \pmod{35}$ is equivalent to the pair of statements $y \equiv 0 \pmod{5}$ and $y \equiv 0 \pmod{7}$. The latter two are easier. To determine those x for which $f(x) \equiv 0 \pmod{5}$, we just try out the possibilities $x = 0, 1, 2, 3, 4$; we find that $f(x) \equiv 0 \pmod{5}$ if and only if $x \equiv 3 \pmod{5}$. To determine those x for which $f(x) \equiv 0 \pmod{7}$, we try out $x = 0, 1, 2, 3, 4, 5, 6$; we find that $f(x) \equiv 0 \pmod{7}$ if and only if $x \equiv 2 \pmod{7}$. Thus $f(x) \equiv 0 \pmod{35}$ if and only if both $x \equiv 3 \pmod{5}$ and $x \equiv 2 \pmod{7}$. We can use the Chinese Remainder Theorem to see that this is equivalent to $x \equiv 23 \pmod{35}$. Thus $f(x) \equiv 0 \pmod{35}$ if and only if $x \equiv 23 \pmod{35}$.

6.3 Problems

(1) What is

$$9 \times 99 \times 999 \times \cdots \times 999\cdots9 \pmod{1000},$$

where the last number contains 999 9's? That is, determine the remainder of the product when divided by 1000.

(2) Prove that if $n \equiv 7 \pmod{8}$, then n cannot be written as a sum of three squares. (Hint: What are the squares modulo 8?)

(3) Prove that for any nonnegative integer n, $3^n + 13 \cdot 10^n$ is divisible by 7.

(4) Describe all integers x satisfying *all* of the following congruences:

$$x \equiv 2 \pmod 3, \qquad x \equiv 5 \pmod 7,$$
$$x \equiv 3 \pmod 5, \qquad x \equiv 7 \pmod{11}.$$

(The answer should be a congruence class modulo some number, such as "all $x \equiv 4 \pmod 9$," although that is not the correct answer in this case.)

(5) Prove that subtraction is well-defined in $\mathbb{Z}/m\mathbb{Z}$, i.e. if $a \equiv b \pmod m$ and $c \equiv d \pmod m$, then $a - c \equiv b - d \pmod m$.

(6) Prove that a number in base 10 is divisible by 9 if and only if the sum of its digits is divisible by 9. Show that a number $d_n d_{n-1} \cdots d_1$ (in terms of its digits) is divisible by 11 if and only if $d_1 - d_2 + d_3 - \cdots + (-1)^{n-1} d_n$ is divisible by 11. Devise, with proof, a (somewhat more involved) test to determine if a number is divisible by 7 or 13.

(7) Let $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ be a polynomial with integer coefficients. Show that if, for some integer $m > 0$, there are m consecutive integers $r, r+1, \ldots, r+m-1$ such that $f(r+i)$ is divisible by m for $0 \le i \le m-1$, then $f(x)$ is divisible by m for every integer x.

(8) A lattice point $(x, y) \in \mathbb{Z}^2$ (i.e. a point in $\mathbb{R}^2$ such that both of its coordinates are integers) is said to be *blocked* if $\gcd(x, y) > 1$. Show that, for any positive integer n, there is an $n \times n$ square consisting entirely of blocked points, i.e. there exist positive integers a and b such that all the points $(a+i, b+j)$ with $0 \le i, j \le n-1$ are blocked. For instance, the four points $(14, 20), (14, 21), (15, 20), (15, 21)$ form a 2×2 square of blocked points.

*, 9, 9x99, 9x99x999, etc. 10/27/26 to 11/1

T+P Week 6: Modular Arithmetic Alexander

6.1: Some iterations* of this mod 1,000:

9, 891, 109, 9891, 109, 9891 ...

Hey, it repeats! Even places (indices) > 2 are 9891, and 109 vice versa. The 999th iteration is an odd index, so it's $\equiv 109 \pmod{1000}$.

6.2: Squares mod 8 are congruent to $1^2, 2^2, 3^2, 4^2, 5^2, 6^2$, or $7^2 \bmod 8$, which are $1, 4, 1, 0, 1, 4, 1 \bmod 8$. The squares can only be 0, 1, or 4, mod 8, and no combination of 0's, 1's, and 4's (when there are 3 of them) are congruent to 7 mod 8 (try $4+1+1=6$, $1+1+0=2$, $4+4+4 \equiv 4$...) ■

6.3: We reduce this mod 7. $13 \equiv 6 \bmod 7$ and $10 \equiv 3 \bmod 7$, so $3^n + 13 \cdot 10^n \equiv 3^n + 6 \cdot 3^n = 7 \cdot 3^n \bmod 7$. That's just $(7 \cdot 3^n) \equiv 0 \bmod 7$! ■

6.4: Going down the list: (All calculations were done by calculator)

$x \equiv 2 \bmod 3$ and $x \equiv 3 \bmod 5$ means $x \equiv 8 \bmod 15$

$x \equiv 8 \bmod 15$ and $x \equiv 5 \bmod 7$ means $x \equiv 68 \bmod 105$.

$x \equiv 68 \bmod 105$ and $x \equiv 7 \bmod 11$ means $\boxed{x \equiv 488 \bmod 1155}$ ■

6.5: Suppose $a-b=km$ and $c-d=\ell m$.
If $a-c \equiv b-d \bmod m$, $(a-c)-(b-d)=a-c-b+d$
$=(a-b)-(c-d)=km-\ell m=(k-\ell)m$. Thus,
subtraction is well-defined in congruences. ∎

6.6: A number $d_n d_{n-1} \ldots d_1 d_0 = d_{n-1} \cdot 10^{n-1} + \ldots + d_0 \cdot 10^0$.
$10 \equiv 1 \bmod 9$, $10 \equiv -1 \bmod 11$. When reduced, said number becomes $d_{n-1} + d_{n-2} \ldots + d_0$ in the 9 case or $\pm d_{n-1} \pm d_{n-2} \pm \ldots \pm d_0$ with alternating + and − in the 11 case. (Then repeat until mult. 9/11)

Part 2:

Note that $1001 = 7 \cdot 11 \cdot 13$, and divisibility rules satisfy those of their factors as well. 1001's divisibility rule is like 11, but it's $\underline{1000} \equiv -1 \bmod 1001$, so it's three digits (every thousand) alternating sum. So 1001 satisfies 7, 11, and 13's rules. So $1000 \equiv -1 \pmod{7, 11, 13}$. Thus, 7 and 13 have the same 3-block rule as 1001.

*Residue is the fancy proper word.

6.7: Lemma: If $x \equiv y \bmod m$, $P(x) \equiv P(y) \bmod m$ for some polynomial (integer coefficients) $P(x)$.

Did I do something wrong? This was REALLY confusing

Proof. Polynomials are of the form $a_k x^k + a_{k-1} x^{k-1} \ldots + a_1 x^1 + a_0$. For addition and multiplication, P is congruent with other inputs. For exponentation: say $by^k \equiv P(x) \bmod m$. Say that $y \equiv x \bmod m$, so $y = x + km$ for some k. Thus:

$b \cdot (x+km)^k \equiv P(x) \equiv bx^k \equiv (x+km)^k = P(x) \equiv ax^k$.

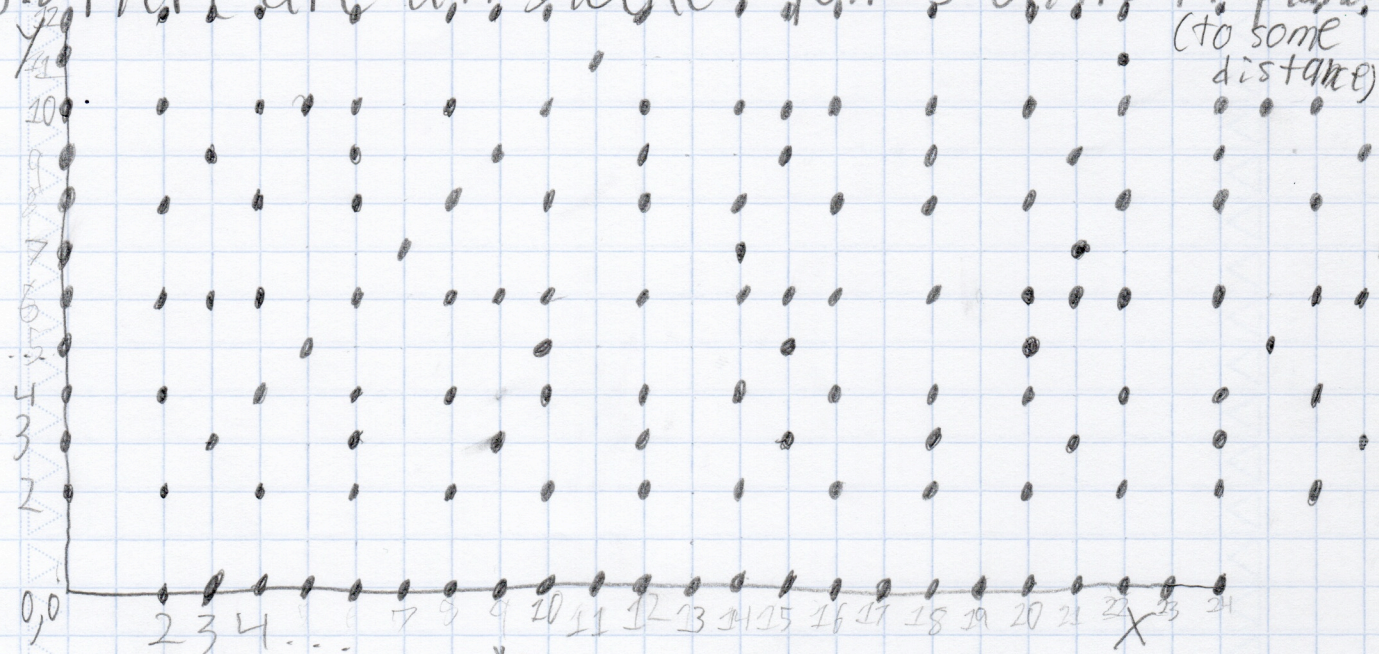
Of course, these are congruent (same exponent).

Thus, if $x \equiv y$ then $P(x) \equiv P(y) \bmod m$.

Proof of 7. Each number in the list $r, r+1 \ldots, r+(m-1)$ has a different remainder* mod m, and by the Lemma $P(x) \equiv P(y) \bmod m$ if $x \equiv y$. Make $x = r$; term, y = remainder when div. m of x. If $x \equiv y$, then each $r+i$ that become $f(r+i) \equiv 0 \bmod m$ occupy a remainder it's congruent to. Thus, $f(r+i) \equiv f(y) \bmod m$, and that remainder y carries for all numbers beyond $r+m$.

* I meant to say squares. YES, there's a pattern.

6.8: Here are all blocked points on the ++ plane: (to some distance)



No pattern*... Let's do it algebraically. Say that $P_{(i,j)}$ represents a prime. We want to find an x such that $x \equiv i_1 \bmod P_{(i_1,j)}$, $x \equiv i_2 \bmod P_{(i_2,j)}$, ... $x \equiv i_n \bmod P_{(i_n,j)}$. According to the CRT, an x exists such that $x \equiv i_0 \bmod P_{(i_1,j)} \cdot \ldots \cdot P_{(i_n,j)}$. The same process can be repeated, but for y and j instead of x and i. The CRT has now shown that an (x,y) exists that (x+i, y+j) is divisible by a prime P, thus creating an n×n square when $0 \le i,j \le n-1$. ∎